Grade: Check Plus

# Assignment 2

| Price (x) | Probability f(x) | x*f(x) | $(x-\mu)^2 * f(x)$ |
|---|---|---|---|
| 20 | 0.05 | 1 | 1.463405 |
| 21 | 0.06 | 1.26 | 1.166886 |
| 22 | 0.07 | 1.54 | 0.813967 |
| 23 | 0.08 | 1.84 | 0.464648 |
| 24 | 0.09 | 2.16 | 0.178929 |
| 25 | 0.12 | 3 | 0.020172 |
| 26 | 0.15 | 3.9 | 0.052215 |
| 27 | 0.13 | 3.51 | 0.328653 |
| 28 | 0.1 | 2.8 | 0.67081 |
| 29 | 0.1 | 2.9 | 1.28881 |
| 30 | 0.05 | 1.5 | 1.053405 |
| | | 25.41 | 7.5019 |

✓ Q1

1 μ = 25.41
standard deviation = 2.74

2a ✓ Q2a
P (25.41 - 2.74 ≤ A ≤ 25.41 + 2.74) =
P (22.67 ≤ A ≤ 28.15) =
P (23 ≤ A ≤ 28) =
f(23) + f(24) + f(25) + f(26) + f(27) + f(28) ... OR ... 1 - (f(30) + f(29) + f(22) + f(21) + f(20))
.67 or 67%

2b ✓ Q2b
P (25.41 - 2.74*2 ≤ A ≤ 25.41 + 2.74*2) =
P (25.41 - 5.48 ≤ A ≤ 25.41 + 5.48) =
P (19.93 ≤ A ≤ 30.89) =
f(20) + f(21) + f(22) + f(23) + f(24) + f(25) + f(26) + f(27) + f(28) + f(29) + f(30) =
1 or 100%

3

| Option value v | Probability P(v) |
|---|---|
| 0 | 0.47 |
| 1 | 0.15 |
| 2 | 0.13 |
| 3 | 0.1 |
| 4 | 0.1 |
| 5 | 0.05 |

✓ Q3

✓ Q4

4 f(26) + f(27) + f(28) + f(29) + f(30) =
.53 or 53%

5

| Option value v | Probability P(v) | x*P(x) | | $(x-\mu)^2 * P(x)$ |
|---|---|---|---|---|
| 0 | 0.47 | | 0 | 0.869312 |
| 1 | 0.15 | | 0.15 | 0.01944 |
| 2 | 0.13 | ✓ Q5 | 0.26 | 0.053248 |
| 3 | 0.1 | | 0.3 | 0.26896 |
| 4 | 0.1 | | 0.4 | 0.69696 |
| 5 | 0.05 | | 0.25 | 0.66248 |
| | | | 1.36 | 2.5704 |

**.47 is derived from adding P(x) for f(20) through f(25)*

1.603246706

μ = 1.36

standard deviation = 1.6

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